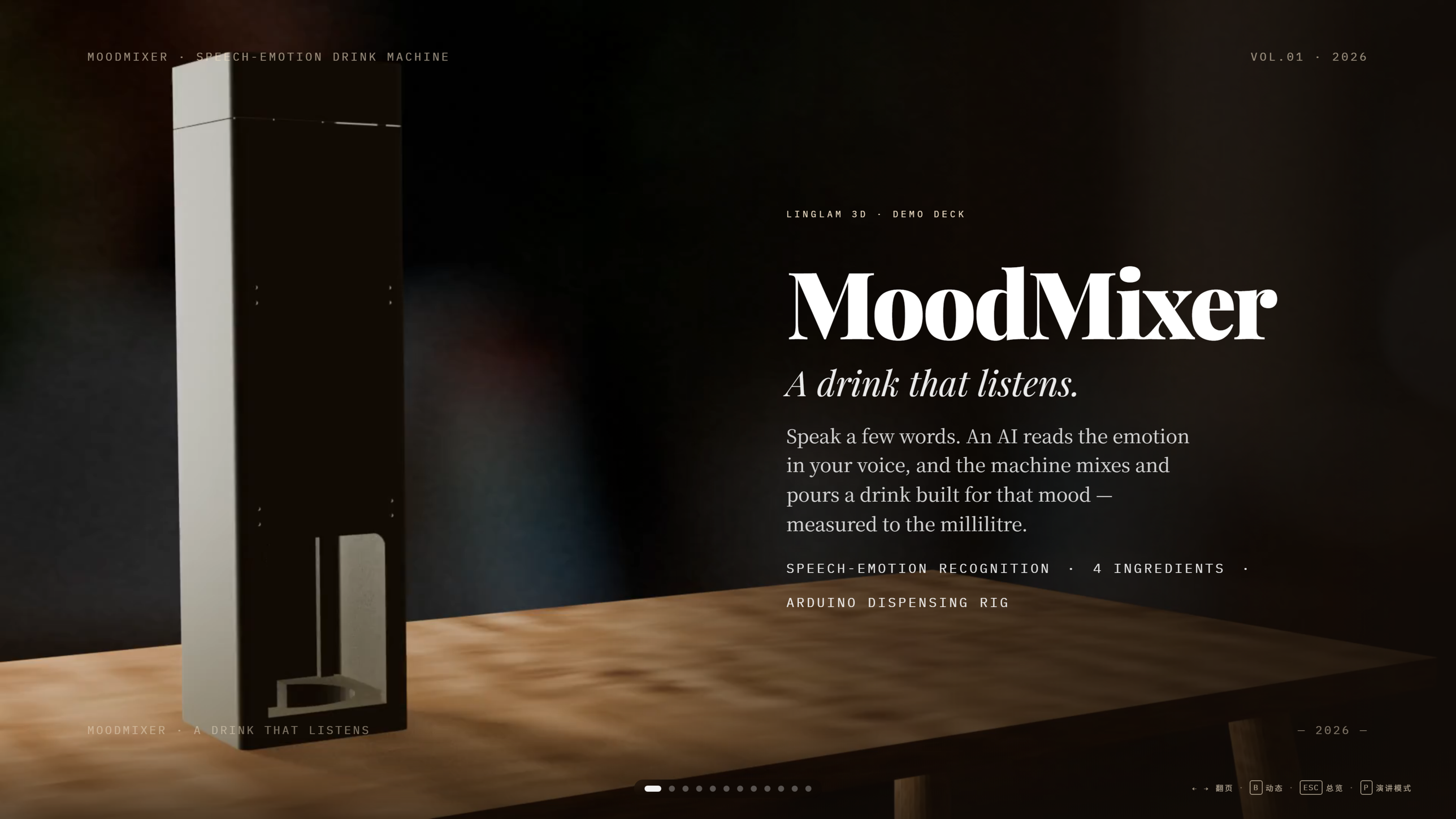


LINGLAM 3D · DEMO DECK

MoodMixer

A drink that listens.

Speak a few words. An AI reads the emotion in your voice, and the machine mixes and pours a drink built for that mood — measured to the millilitre.

SPEECH-EMOTION RECOGNITION · 4 INGREDIENTS · ARDUINO DISPENSING RIG

THE PREMISE

A menu asks what you want. *It never asks how you feel.*

MoodMixer closes that gap. Your voice already carries a mood — the machine just listens, then turns it into a recipe.

LISTEN FIRST · THEN POUR · NO MENU

THE IDEA

It hears how you feel.

You don't tap a menu. You say a few words. A speech-emotion model reads the tone — not the words — and picks one of four moods.

NEUTRAL HAPPY ANGRY SAD

> "The input is a voice. The output is a drink."
>
> MOODMIXER · END TO END



HOW IT WORKS · 5 STEPS

From voice to cup

A browser records, a FastAPI service classifies, and a recipe comes back in millilitres — ready for the Arduino.

SIGNAL CHAIN · SENSE → DECIDE → POUR

01

Speak

Record with the browser mic, or upload a voice clip.

02

Encode

Decode with soundfile, resample to 16 kHz mono.

03

Classify

wav2vec2 returns the mood and confidence scores.

04

Recommend

The mood maps to a drink recipe in millilitres.

05

Pour

The Arduino opens valves; the load cell verifies.

Handoff

The recipe returns as a small object of millilitres — the only contract between the AI and the machine.

RECIPE.JSON

← → 翻页 · B 动态 · ESC 总览 · P 演讲模式

FOUR INGREDIENTS

Four ingredients. Four moods. One cup.

Jasmine green tea, Ceylon black tea, fresh milk, and sugarcane syrup — combined to the millilitre for each emotion.

茉莉綠茶 · 錫蘭紅茶 · 鮮奶 · 蔗糖漿

FOUR MOODS, FOUR RECIPES

Mood in, recipe out.

HAPPY

Caramel Milk Tea

焦糖鮮奶茶

CEYLON 180
MILK 100 · SYRUP 20

SAD

Comfort Milk Green

暖心鮮奶綠

JASMINE 160 · MILK 130
SYRUP 10

ANGRY

Calming Jasmine Green

清心茉莉綠

JASMINE 300
NOTHING ELSE

NEUTRAL

Classic Ceylon Black

經典錫蘭紅

CEYLON 300
NOTHING ELSE



FOUR BASE INGREDIENTS · ONE COUNTER

PRECISION

It doesn't guess. *It weighs.*

A 5 kg load cell closes the loop. Each valve opens until the target millilitres are in the cup — not a splash more.

HX711 · 5 KG LOAD CELL · CLOSED-LOOP POURING

UNDER THE HOOD

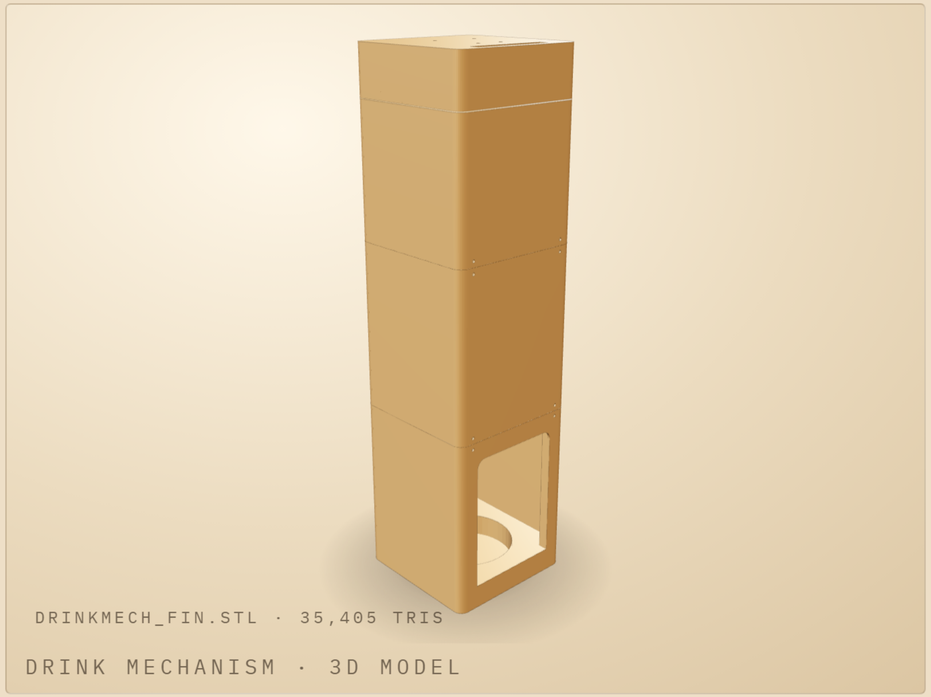
The rig.

An Arduino Uno drives four food-grade solenoid valves. A 3D-printed drink mechanism sits at the heart, the turntable moves the cup, and the load cell keeps every pour honest.

Arduino Uno	CTRL	4× 12V solenoid valves	POUR
IRLZ44N + flyback diodes	DRIVE	HX711 + 5 kg load cell	SENSE
NEMA17 + A4988 turntable	MOVE	12V 5A PSU + LM2596	POWER

4 VALVES · 5 KG LOAD CELL · ~¥550–700 BOM

3D MODEL · 145 × 544 × 145 MM · STL



DRINKMECH_FIN.STL · 35,405 TRIS

DRINK MECHANISM · 3D MODEL

← → 翻页 · B 动态 · ESC 总览 · P 演讲模式

SOFTWARE

Four layers, one pour.

Model	*superb/wav2vec2-base-superb-er* — a pre-trained wav2vec2 audio transformer, out of the box.	4 EMOTIONS
Backend	Python 3.13 • FastAPI • Uvicorn. One `POST /predict` endpoint.	/PREDICT
Frontend	Single-page vanilla HTML / CSS / JS — record in the browser or upload a clip.	MIC + UPLOAD
Firmware	Arduino maps millilitres to valve-open time; the load cell confirms the pour.	ML → MS

Audio is decoded with *soundfile* and resampled to 16 kHz mono with *scipy* — exactly the input the model expects.

THE PITCH

Why a mood, not a menu?

01

Emotion-aware

A drink matched to how you actually feel — not what you tap.

02

Zero-training

A pre-trained state-of-the-art model, deployed with no dataset to collect.

03

Hands-free

Speak and it pours. No app, no menu, no screen to touch.

04

Modular & cheap

Commodity parts and an open recipe system — easy to extend.

05

Novel experience

A strong pull for cafes, pop-ups, events and vending.

06

Open recipes

New ingredients and emotions are a config change, not a rebuild.

FROM PROTOTYPE TO POUR

The software is done. The pour is next.

Classification and recipes already work end to end. The remaining work is turning millilitres into precise, repeatable pours.

NOW · 01

Wire the API

Map each recipe's millilitres to valve-open time and pour from the Arduino.

NEXT · 02

Close the loop

Use load-cell feedback to cut each valve at the target weight.

LATER · 03

More moods

Expand beyond four emotions and add ingredients to the recipe set.

LATER · 04

Calibration

A self-tune mode for flow rates, so every bottle pours the same.

The first reliable pour is the milestone.

MOODMIXER

Speak. Sip. Repeat.

A drink matched to how you feel — an off-the-shelf model, four ingredients, and a load cell.

GITHUB.COM/PERCY-THE-TRAIN/MOODMIXER · LINGLAM 3D · 2026